BYZANTINE-RESILIENT FEDERATED LEARNING FOR TACTICAL UAV SWARMS OVER LOSSY WIRELESS LINKS

Project report submitted to the
National Institute of Technology Warangal
For award of the degree

of

Bachelor of Technology

in Electronics and Communication Engineering

by

Nasika Sasank	**23ECB0A03**
Aaron George	**23ECB0F06**
Sachin Sreekumar Nair	**23ECB0F11**

Under the guidance of

Dr. Chayan Bhar



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

November 2026

APPROVAL OF THE VIVA-VOCE BOARD

DD/MM/YYYY

Certified that the project report entitled BYZANTINE-RESILIENT FEDERATED LEARNING FOR TACTICAL UAV SWARMS OVER LOSSY WIRELESS LINKS submitted by NASIKA SASANK (23ECB0A03), AARON GEORGE (23ECB0F06), SACHIN SREEKUMAR NAIR (23ECB0F11) to the National Institute of Technology, Warangal, for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering has been accepted by the examiners and that the students have successfully defended the work in the viva-voce examination held today.

(Examiner) (Examiner) (Examiner)

(Supervisor) (Head of the Department)

(Chairman)

(The certificate is to be printed on the Departmental Letter-Head)

CERTIFICATE

This is to certify that the project report entitled **Byzantine-Resilient Federated Learning for Tactical UAV Swarms over Lossy Wireless Links**, submitted by **Nasika Sasank** (23ECB0A03), **Aaron George** (23ECB0F06), **Sachin Sreekumar Nair** (23ECB0F11) to the National Institute of Technology Warangal, is a record of bona fide work carried out under my supervision in the Department of Electronics and Communication Engineering during the academic year 2026–2027, and we consider it worthy of consideration for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering of the Institute.

Dr. Chayan Bhar	Prof. Ravi Kumar Jatoth
Supervisor	Head of the Department

Date:

DECLARATION

We certify that

a. The work contained in this report is original and has been done by us under the general supervision of our supervisor.

b. The work has not been submitted to any other Institute for any degree or diploma.

c. We have followed the guidelines provided by the Institute in writing the report.

d. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references.

e. Whenever we have quoted written materials from other sources, we have put them under quotation marks and given due credit to the sources by citing them and giving required details in the references.

Nasika Sasank	Aaron George	Sachin Sreekumar Nair
23ECB0A03	23ECB0F06	23ECB0F11

Signature of the Students

Acknowledgements

We thank our supervisor, Dr. Chayan Bhar, for steering this project towards realistic wireless conditions and fair, like-for-like comparisons, and for asking us to state our findings with the evidence behind them. We thank the Head of the Department, Prof. Ravi Kumar Jatoth, and the faculty of the Department of Electronics and Communication Engineering for their support. We also thank the creators of the DeepSense 6G dataset for making their UAV measurements available.

Nasika Sasank

Aaron George

Sachin Sreekumar Nair

Abstract

Swarms of unmanned aerial vehicles (UAVs) can learn a shared perception model with federated learning (FL): each UAV trains on the images it collects and sends only a model update to a ground base station. In a tactical setting two problems arise together: air-to-ground links lose packets, and some UAVs may be captured and made to send poisoned updates. Byzantine-robust aggregation rules were designed for complete updates, while work on FL over erroneous links, including our base paper (Shirvanimoghaddam et al., IEEE Communications Letters, 2022), assumes honest clients. This project studies both problems in one system.

We built a PyTorch simulation of hierarchical FL over a UAV swarm. Ten UAVs holding non-IID, aerial-transformed CIFAR-10 images are served by three base stations, which aggregate locally and synchronize every five rounds. Packet erasures follow the relative power variation of a measured drone trace (DeepSense 6G Scenario 23), replayed around assumed reference signal-to-noise ratio anchors with virtual neighbouring base stations. Three UAVs mount sign-flip, ALIE-style or inner-product-manipulation attacks. We designed CCRA, a reputation-based aggregator that compares each update with a reference computed on a small root dataset and can use the channel in two ways: by ignoring lost coordinates (erasure masking) and by letting link telemetry slow how fast a UAV regains trust. The base paper's two erasure rules, fresh-only averaging and reuse of old updates, are evaluated in a separate 240-run follow-up.

A FedAvg baseline reached 62.6% (IID) and 60.2% (non-IID) accuracy, and a preliminary single-seed study looked promising for CCRA. To find which part of CCRA was responsible, a prespecified, paired study of 720 runs crossed six aggregators, a clean setting and three attacks, three channel conditions and 10 seeds. The primary telemetry stress interaction was -2.7 percentage points (95% paired-seed bootstrap interval $[-3.9, -1.5]$),

beyond the prespecified −1 point threshold. Under attack at the 5 dB anchor, erasure masking improved accuracy by 4.2 points ([2.2, 5.9]), whereas adding telemetry changed it by -3.5 points ([−5.1, −2.1]), accompanied by a rise in honest-update rejection from 13.8% to 41.2%. A constant-evidence drift analysis suggests why: when rewards are slowed but penalties are not, the equilibrium reputation of an honest UAV on a weak link moves toward the lower tail of its scores. None of the four root-referenced aggregators diverged, whereas FedAvg and the median diverged in 41 and 26 of their 120 runs.

Against the base paper, mask-only aggregation was 19.2 points more accurate than its better rule, fresh-only, under attack at 5 dB ([15.9, 23.0]), at a cost of -2.1 points without attack ([−2.9, −1.4]). The base paper's rules diverged in 41/120 and 42/120 runs, including every sign-flip run, and fresh-only beat reuse-old by 5.39 points on clean stressed links. This follow-up ran on Windows GPU workers; on the ideal channel it matches the A100 study within 1.1 points per run.

The useful channel-aware component is therefore erasure masking, and the reward rule must be redesigned before telemetry can help. The results are written up as an IEEE letter. Phase II will revise the reward rule and test it on fresh seeds, higher-mobility traces and adaptive attackers, including hardware-controlled confirmation of the follow-up contrasts.

Index Terms : Federated learning, UAV swarm, Byzantine robustness, packet erasure, hierarchical aggregation, reputation.

Contents

Certificate of Approval ii

Certificate iii

Declaration iv

Acknowledgements v

Abstract vi

List of Figures xii

List of Tables xiii

List of Abbreviations xiv

List of Symbols xv

1 Introduction 1

1.1 Background 1

1.2 Problem Statement 2

1.3 Aim and Scope 2

1.4 Objectives 3

1.5 Contributions 3

1.6 Organization of the Report 4

2 Literature Review **5**

2.1 Federated Learning 5

2.2 Byzantine-Robust Aggregation 5

2.3 Federated Learning over Unreliable Wireless Links 7

2.4 Federated Learning for UAV Swarms 7

2.5 Research Gap 8

3 Theoretical Study **10**

3.1 Hierarchical Federated Learning for a UAV Swarm 10

3.1.1 Dataset and Model 11

3.2 Wireless Uplink Models 12

3.2.1 Synthetic Jammed Channel 12

3.2.2 Measured-Trace Channel 12

3.2.3 Packet Erasures 13

3.3 Threat Model 14

3.4 Aggregation Rules 15

3.4.1 Baselines 15

3.4.2 Context-Calibrated Robust Aggregation (CCRA) 16

3.4.3 Ablation Variants 17

3.4.4 Analysis of the Reputation Update 17

4 Experimental Study **19**

4.1 Implementation and Verification 19

4.2 Experimental Protocol 20

4.2.1 Preliminary Study 20
4.2.2 Controlled Ablation Study 20
4.2.3 Comparison with the Base Paper 21
4.3 FedAvg Baseline Results 21
4.4 Preliminary Study Results 22
4.5 Controlled Ablation Study Results 23
4.5.1 Stability and Overall Accuracy 24
4.5.2 Which Kind of Channel Awareness Helps 25
4.5.3 Mechanism: Who the Gate Rejects 26
4.5.4 Comparison with FLTrust 27
4.6 Completed Base-Paper Comparison 28
4.6.1 Our Method against the Base Paper 29
4.7 Discussion 30
4.7.1 What the Results Show 30
4.7.2 Why a Negative Result Is Useful 30
4.7.3 Threats to Validity 30

5 Conclusions and Future Scope of Study 32

5.1 Summary of the Work 32
5.2 Conclusions 32
5.3 Scope for Future Work (Phase II) 33

References 35

Appendix-A 38

Appendix-B **39**

List of Figures

3.1 Hierarchical FL over a UAV swarm. Each base station aggregates the updates of the UAVs it serves, some of which arrive with lost packets; edge models are synchronized every E rounds. 10

3.2 Packet error rate against SNR: the base paper's normal approximation for a 32,768-bit packet at rate 1.58 bits per channel use, the same curve averaged over a 1.7 dB Gaussian SNR spread, and the logistic curve used in this work. 14

4.1 Test accuracy and loss of the FedAvg baseline over 50 rounds. 22

4.2 Preliminary study: final accuracy in the four scenarios. 23

4.3 Preliminary study: aggregation weight given to attackers and to jammed honest UAVs under label flipping on the jammed channel. 23

4.4 Final accuracy under attack (mean over sign flip, ALIE and IPM and 10 seeds). Error bars: 95% seed-bootstrap intervals. Dashed, starred rules are from the separate 240-run follow-up; cross-study rankings are descriptive. . 24

4.5 Paired accuracy differences under attack with 95% bootstrap intervals. The grey band is the prespecified ± 1 point region of practical equivalence. . . . 26

4.6 Gate decisions under attack at 5 dB, as a percentage of received updates, averaged over attacks and seeds. 27

List of Tables

2.1 Coverage of the problem in the reviewed work 9

4.1 FedAvg baseline accuracy after 50 rounds 21

4.2 Preliminary study: final accuracy (%), single seed, 30 rounds. CosAvg is our CosAvg-style baseline, not the published method. 22

4.3 Outcomes over all 720 runs (120 per method). Diverged runs count as 10%. 24

4.4 Mean final accuracy (%) by attack, method and channel. Best value in each block and column in bold. 25

4.5 Separate base-paper follow-up: ten paired seeds per cell. Final accuracy includes numerical divergences at chance (10%). Intervals are exploratory paired seed-bootstrap intervals, with no multiplicity adjustment. 28

A.1 Settings of the controlled study 38

B.1 Repository layout 39

List of Abbreviations

ALIE	A Little Is Enough (model-poisoning attack)
BS	Base station
CCRA	Context-Calibrated Robust Aggregation (the aggregator studied here)
CI	Confidence interval
CNN	Convolutional neural network
CosAvg	Cosine-similarity weighted averaging
EMA	Exponential moving average
EW	Electronic warfare
FedAvg	Federated averaging
FL	Federated learning
HFL	Hierarchical federated learning
IID	Independent and identically distributed
IPM	Inner-product manipulation (model-poisoning attack)
PDR	Packet delivery ratio
PER	Packet error rate
pp	Percentage points
RMS	Root mean square
SGD	Stochastic gradient descent
SINR	Signal-to-interference-plus-noise ratio
SNR	Signal-to-noise ratio
TEE	Trusted execution environment
UAV	Unmanned aerial vehicle

List of Symbols

K	Number of UAVs (clients)
B	Number of base stations
E	Edge rounds between global synchronizations
P	Number of model parameters
L	Parameters per packet
f	Number of Byzantine UAVs
$\mathcal{D}_k$, n_k	Local dataset of UAV k and its size
$\mathcal{D}_0$	Trusted root dataset held by each base station
$\mathbf{w}^t$, $\mathbf{w}_b^t$	Global model and edge model of BS b in round t
$\boldsymbol{\Delta}_k^t$	Model update sent by UAV k in round t
$\mathbf{m}_k$	Delivery mask: 1 where the packet carrying a coordinate arrived
$\mathbf{r}$	Reference update computed on $\mathcal{D}_0$
γ_k, $\bar{\gamma}$	Uplink SNR of UAV k; reference SNR of the trace
p_k	Packet error probability of UAV k
a, γ_{th}	Slope and threshold of the packet-error curve
s_k	Norm-clipping factor
c_k	Masked cosine similarity to the reference update
b_k	Behaviour score in $[0, 1]$
T_k	Reputation (trust) of UAV k
e_k	Link evidence in $[0, 1]$ that slows reputation rewards
β, τ	Reputation memory and exclusion gate
ω_k	Aggregation weight of UAV k
α	Dirichlet concentration of the non-IID data split
λ, z, ϵ	Strengths of the sign-flip, ALIE and IPM attacks

Chapter 1

Introduction

1.1 Background

Small unmanned aerial vehicles (UAVs) are now used in groups, or swarms, for surveillance, search and rescue, disaster assessment and border patrol. Each UAV carries a camera and collects images of the area it covers. Training across the datasets of several UAVs may improve generalization compared with training on a single UAV's data [McMahan et al., 2017], but sending raw imagery to a ground station is costly in bandwidth and exposes sensitive data if a link is intercepted.

Federated learning (FL) addresses this by moving training to the data. Each UAV trains a copy of the model on its own images and uploads only the resulting model update. A server (here, a ground base station) combines the updates into a new global model and sends it back. The raw images remain on the UAV, and an update is usually much smaller than the data it was computed from. Updates can still leak some information about the data, so FL alone is not a privacy guarantee.

A tactical swarm, however, breaks two assumptions that standard FL relies on.

1. **Links are unreliable.** Air-to-ground channels fade as the UAV moves, hand over between base stations, and may be deliberately jammed. Parts of an update are lost, and some updates miss the round deadline entirely.

2. **Some clients may be hostile.** A captured or spoofed UAV can follow the protocol while sending a poisoned update designed to damage the global model. Such clients

are called *Byzantine*.

Most Byzantine-robust aggregation rules assume that every update arrives intact, and most work on FL over imperfect channels assumes that every client is honest. A swarm needs both at once.

1.2 Problem Statement

Given a UAV swarm that trains a shared image classifier with hierarchical FL over lossy air-to-ground links, in which a minority of UAVs are Byzantine, design and evaluate an aggregation rule that (i) keeps the global model accurate and stable, (ii) does not penalize honest UAVs simply because their links are poor, and (iii) can be compared fairly against established rules under identical conditions.

1.3 Aim and Scope

The aim of this project is to determine *which* kind of channel awareness actually helps a robust aggregator in a UAV swarm, and to quantify the effect with a controlled, reproducible experiment. Its scope in Phase I is:

- a simulation of hierarchical FL with up to three base stations, UAV association and handover;
- packet-level uplink losses from a synthetic jammed channel and from a measured UAV received-power trace;
- five supported attacks (label flip, sign flip, Gaussian noise, ALIE and IPM), of which the preliminary study uses label flip and the controlled study the three model-poisoning attacks sign flip, ALIE and IPM;
- six aggregation rules evaluated on the same data, models, attackers and channel realizations, and the base paper's two erasure rules for a separate comparison.

Hardware flight tests, real radio transmission and attacks that forge link telemetry are outside the Phase I scope.

1.4 Objectives

1. Build an FL baseline for a UAV swarm on aerial-transformed CIFAR-10 and measure its accuracy under IID and non-IID data.
2. Model a lossy uplink in which lost packets erase parts of an update, and add Byzantine attackers.
3. Design a reputation-based aggregator (CCRA) that uses a trusted reference update, and give it two forms of channel awareness: erasure masking and telemetry-weighted reputation.
4. Separate the effect of each component with a prespecified, multi-seed ablation study, and report the results with honest uncertainty estimates.

1.5 Contributions

- An open, tested PyTorch framework for hierarchical UAV-swarm FL with trace-driven packet erasures, handover and five attacks, with automated tests.
- A preliminary study, then a controlled 720-run study whose analysis plan and decision threshold were fixed before the runs.
- The finding that erasure masking improves accuracy under attack on weak links by 4.2 percentage points, while telemetry-weighted reputation lowers it, together with an analysis of the reputation rule that suggests a mechanism consistent with this result.
- A completed 240-run comparison of the base paper's two erasure rules on matched seeds, attackers and channels, with validated checkpoint recovery and an explicit separation from the prespecified ablation.
- A four-page manuscript in IEEE letter format that reports these results.

1.6 Organization of the Report

Chapter 2 reviews federated learning, Byzantine-robust aggregation and FL over unreliable wireless links, and identifies the gap this project addresses. Chapter 3 presents the theoretical study: the system and threat models, the aggregation rules and an analysis of the reputation update. Chapter 4 presents the experimental study: implementation, protocol, the baseline, the preliminary study and the controlled study, and a discussion of what they show. Chapter 5 states the conclusions and the plan for Phase II. The appendices, after the references, list the experimental settings and the steps needed to reproduce every result.

Chapter 2

Literature Review

2.1 Federated Learning

McMahan et al. [2017] introduced federated averaging (FedAvg). In each round the server sends the global model to a set of clients, each client runs a few epochs of stochastic gradient descent on its local data, and the server replaces the global model by the average of the returned models, weighted by local dataset size. FedAvg works well when client data are independent and identically distributed (IID). When each client sees a skewed mix of classes, local models drift apart and accuracy falls. Hsu et al. [2019] proposed sampling each client's class proportions from a Dirichlet distribution with concentration α, which has become the standard way to control the degree of non-IID skew in experiments; we use $\alpha = 0.5$.

Hierarchical FL adds an intermediate layer of edge servers. Clients send updates to a nearby edge server, which aggregates them frequently, and the edge servers synchronize with each other less often [Abad et al., 2020]. This matches a UAV swarm served by several ground base stations, each covering part of the operating area.

2.2 Byzantine-Robust Aggregation

A single malicious client can move the FedAvg average arbitrarily far, so a large body of work replaces the mean by a robust statistic.

- **Krum** [Blanchard et al., 2017] selects the update whose summed distance to its nearest neighbours is smallest. It requires $n \geq 2f + 3$ clients to tolerate f attackers.
- **Coordinate-wise median and trimmed mean** [Yin et al., 2018] take the median, or a trimmed mean, of each parameter separately, and come with statistical error guarantees.
- **FLTrust** [Cao et al., 2021] gives the server a small clean root dataset. The server computes its own update on that data each round, scores each client by the (clipped at zero) cosine similarity to it, and rescales every client update to the norm of the server update before averaging.
- **Geometric median** (RFA) [Pillutla et al., 2022] aggregates with an approximate geometric median, which differs from the coordinate-wise median.

Non-IID data make honest updates look different from each other, which a robust rule can mistake for an attack; bucketing [Karimireddy et al., 2022] adapts robust rules to heterogeneous data. Attacks that accumulate small changes over rounds are countered by using history, through worker momentum and iterative clipping [Karimireddy et al., 2021]; this is related to, but different from, keeping a reputation per client. These rules are evaluated against attacks of increasing strength. A sign-flip attack sends a scaled negative of an honest update. ALIE ("A Little Is Enough") [Baruch et al., 2019] shifts each coordinate by a small multiple of the honest standard deviation, which is enough to evade distance-based rules. Inner-product manipulation (IPM) [Xie et al., 2019] sends a negative multiple of the honest mean so that the aggregate's inner product with the true gradient becomes negative.

Reputation-based defences keep a trust score for each client across rounds, so that an attacker caught once stays down-weighted. They suit swarms, whose members persist for a whole mission, but recent work shows that adaptive attackers can learn to evade them: Liu et al. [2026] model the attacker's choice between benign and malicious rounds as a partially observable Markov decision process over the server's hidden trust score, and suppress accuracy under reputation defences that stop static attacks.

2.3 Federated Learning over Unreliable Wireless Links

In wireless FL the update itself travels over a noisy channel. Shirvanimoghaddam et al. [2022], the base paper of this project, model each device-to-server link as a packet erasure channel: a device's whole update is either erased or received, with an erasure probability obtained from the normal approximation for short-packet transmission. They compare two server rules, averaging only the fresh updates received in a round or reusing a device's last received parameters when its fresh update is missing. They prove that, for convex and smooth local losses, the reuse rule converges to the same global parameters as error-free FL, and observe that with uniformly distributed data the fresh-only rule can converge faster. Their model has no adversaries. We adopt the erasure view at packet granularity, so that an update can arrive partially, and use both of their rules as baselines (Section 3.4.1). Barkatsa et al. [2024] consider malicious clients that jam the non-orthogonal multiple access (NOMA) uplink of legitimate ones, and let every client choose its transmit power in a Bayesian game with incomplete information about the other clients' types. Li et al. [2022] combine communication efficiency (1-bit compressed sensing of sign-based gradients), differential privacy and Byzantine robustness, but do not model lost packets. Byzantine robustness has also been studied for analog over-the-air aggregation, in which clients transmit simultaneously and the channel itself computes a sum: Sifaou and Li [2022] aggregate robustly across random groups, Yao et al. [2025] combine Byzantine identification, adaptive clustering and aggregation weights, and Fan et al. [2026] use UAVs to collect gradients over the air from heterogeneous workers under Byzantine attack. In those systems the server cannot observe individual updates, whereas our UAVs send individually observable digital updates that may lose packets.

2.4 Federated Learning for UAV Swarms

Xiao et al. [2021] propose fully decentralized FL for on-board missions of a UAV swarm: UAVs exchange models with neighbours over a time-varying graph instead of through a server, using an inexact stochastic parallel random-walk ADMM, and the framework is demonstrated on beamforming design. It considers neither adversarial UAVs nor

packet loss. Liao et al. [2023] propose UAV-assisted FL in which the UAV aggregates inside a trusted execution environment (TEE), using a robust rule they call CosAvg: each client is scored by the sum of its pairwise cosine similarities with the others, and the top-ranked updates are averaged. Our preliminary study used a simpler CosAvg-style baseline that weights each update by its cosine similarity to the mean update; it is not an exact reproduction. Realistic UAV link data have recently become available: the DeepSense 6G dataset [Alkhateeb et al., 2023] includes a scenario with a flying UAV and a ground base station [Charan and Alkhateeb, 2025], which we use to drive packet losses in simulation.

2.5 Research Gap

Table 2.1 summarizes the reviewed work; it is a small sample and does not establish that no other work covers these points. Prior work addresses Byzantine robustness in conventional FL, where robust rules were defined for complete update vectors, and in analog over-the-air aggregation, including UAV-assisted architectures. Work on erroneous links, such as the base paper, assumes honest clients. This project studies a different setting: digital packet erasures in hierarchical FL with UAV clients, together with model-poisoning attackers. It is also natural to assume that link telemetry (signal quality, delivery ratio, handover state) should make a robust aggregator *fairer* to honest UAVs on poor links. This project tests that assumption with a paired ablation that separates delivered-coordinate masking from telemetry-weighted reputation.

Table 2.1 Coverage of the problem in the reviewed work. "Poisoned updates" means model- or data-poisoning clients; "digital erasures" means individually received updates that can be lost in whole or in part.

Work	Poisoned updates	Digital erasures	UAV setting	Measured channel
FedAvg [McMahan et al., 2017]	–	–	–	–
Krum, median [Blanchard et al., 2017, Yin et al., 2018]	✓	–	–	–
FLTrust [Cao et al., 2021]	✓	–	–	–
Erroneous links [Shirvanimoghaddam et al., 2022]	–	✓	–	–
Jamming mitigation [Barkatsa et al., 2024]	–[a]	✓	–	–
DP + Byzantine [Li et al., 2022]	✓	–	–	–
Over-the-air robust FL [Sifaou and Li, 2022, Yao et al., 2025]	✓	–[b]	–	–
UAV-assisted over-the-air FL [Fan et al., 2026]	✓	–[b]	✓	–
UAV-swarm FL [Xiao et al., 2021]	–	–	✓	–
TEE + CosAvg [Liao et al., 2023]	✓	–	✓	–
This project	✓	✓	✓	✓[c]

[a]Malicious clients jam the uplink; they do not send poisoned updates. [b]Analog channel impairment of a superposed sum, not digital erasures. [c]Relative power variation of one measured link, with virtual neighbouring BSs and assumed SNR anchors.

Chapter 3

Theoretical Study

This chapter describes the simulated system, the attacker and the aggregation rules compared, and analyses the reputation update. Chapter 4 describes how they were evaluated. All of it is implemented in the project repository; file names are given where they help locate the code.

3.1 Hierarchical Federated Learning for a UAV Swarm

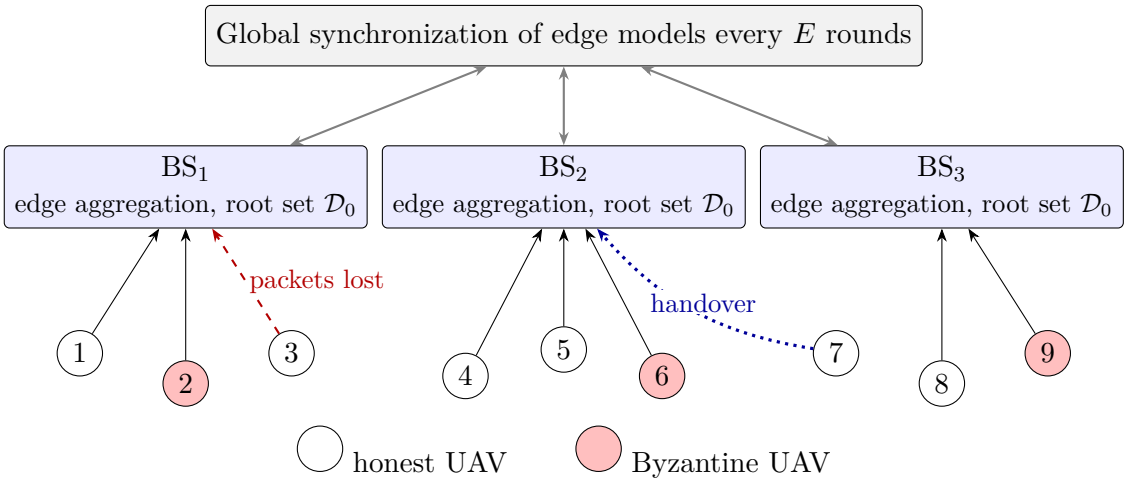


Figure 3.1 Hierarchical FL over a UAV swarm. Each base station aggregates the updates of the UAVs it serves, some of which arrive with lost packets; edge models are synchronized every E rounds.

Figure 3.1 shows the system. K UAVs are served by B ground base stations (BSs). Each UAV k holds a private dataset $\mathcal{D}_k$ of n_k images. In every round t:

1. each UAV attached to BS b starts from that BS's edge model $\mathbf{w}_b^t$ and trains it for

one local epoch of SGD on $\mathcal{D}_k$;

2. it uploads the update $\boldsymbol{\Delta}_k^t$ (new local model minus the edge model), which is split into packets of L parameters;

3. the BS aggregates the updates it receives and sets $\mathbf{w}_b^{t+1} = \mathbf{w}_b^t + \mathrm{Agg}(\cdot)$;

4. every E rounds the edge models are averaged, weighted by the number of training samples held by the UAVs in each cell, and the result is sent back to every BS.

UAVs attach to the BS with the strongest received power. A UAV hands over only when another BS is stronger by a hysteresis margin of 3 dB for two consecutive rounds, which avoids ping-pong handovers (`fl/topology.py`).

3.1.1 Dataset and Model

Images come from CIFAR-10, transformed to imitate some properties of aerial imagery (`fl/data.py`); we call the result aerial-transformed CIFAR-10, since the transformations do not make it genuine UAV imagery. Each image is rotated by a random multiple of 90°, since a downward-looking camera has no preferred orientation, and with probability 0.3 or 0.2 is down-sampled to 24×24 or 16×16 pixels and up-sampled again to imitate higher flight altitudes. Training images are split among the UAVs either uniformly (IID) or with Dirichlet(α) class proportions. In the controlled study of Chapter 4, 500 images (50 per class) are held out as the root set $\mathcal{D}_0$, which is assumed clean and never given to any UAV, and the remaining 49,500 are split among the UAVs. The earlier baseline experiments split all 50,000 images among the UAVs and used no root set.

The classifier, TacticalEdgeCNN, is a small convolutional network with three convolutional layers (16, 32 and 64 channels) and two fully connected layers, 156,074 trainable parameters in total and no normalization layers, chosen so that it could plausibly be trained on board. Without normalization layers the whole model state consists of plain weights, which keeps averaging across UAVs straightforward.

3.2 Wireless Uplink Models

The channel decides, for every packet of every update, whether it arrives (`fl/channel.py`). Two models are used.

3.2.1 Synthetic Jammed Channel

UAVs move by a random walk at 100 m altitude over an 800×800 m area. The received power from UAV k at its serving BS is

$$P_{\mathrm{rx},k} = P_{\mathrm{tx}} - \mathrm{PL}(d_k) + 10\log_{10}|h_k|^2, \qquad \mathrm{PL}(d) = 40 + 10\eta\log_{10} d \ \mathrm{dB}, \tag{3.1}$$

with $P_{\mathrm{tx}} = 20$ dBm, path-loss exponent $\eta = 2.2$, and Rayleigh fading $|h_k|^2 \sim \mathrm{Exp}(1)$. A ground jammer of power 10 dBm targets a sector of the swarm, and the SINR is

$$\gamma_k = P_{\mathrm{rx},k} - 10\log_{10}\big(N_0 + J_k\big), \tag{3.2}$$

where N_0 (-90 dBm) and the received jamming power J_k (zero for UAVs outside the jammed sector) are added as linear powers in mW before conversion to dBm. This jammed channel is used only in the preliminary study; the controlled study of Chapter 4 has no jammer.

3.2.2 Measured-Trace Channel

For realism we replay a measured UAV link from DeepSense 6G Scenario 23 [Alkhateeb et al., 2023, Charan and Alkhateeb, 2025], in which a drone flies around a ground receiver that records 60 GHz beam powers together with position, altitude and speed. The dataset's power values are uncalibrated, so only their variation relative to the median is used:

$$\gamma_k = \bar{\gamma} + \big(P_k^{\mathrm{meas}} - \mathrm{med}\, P^{\mathrm{meas}}\big) - 10\eta\log_{10}\frac{d_{k,b}}{d_{k,1}}, \tag{3.3}$$

where P_k^{meas} is the recorded beam power in dB on the dataset's arbitrary scale, $\bar{\gamma}$ is an assumed reference SNR, and the last term adjusts the measured link to the distance of

a virtual BS b, with distances in metres so that their ratio is dimensionless. Each UAV replays a different flight segment. Two settings are used: $\bar{\gamma} = 15\,\mathrm{dB}$, a good link, and $\bar{\gamma} = 5\,\mathrm{dB}$, a weak link. They replay identical trajectories, so only the link quality differs.

3.2.3 Packet Erasures

In both models, conditional on the current link state, packets are lost independently; all L parameters of a packet share one erasure. Each packet is lost with probability

$$p_k = \frac{1}{1 + \exp\big(a(\gamma_k - \gamma_{\mathrm{th}})\big)}, \tag{3.4}$$

with $a = 1$ and $\gamma_{\mathrm{th}} = 3\,\mathrm{dB}$. An update whose transmission time at the Shannon rate exceeds the 2 s round deadline is dropped entirely. The BS therefore receives $\tilde{\mathbf{\Delta}}_k = \mathbf{m}_k \odot \mathbf{\Delta}_k$, where the mask $\mathbf{m}_k$ is 1 for delivered coordinates and 0 elsewhere. The BS also records link telemetry for each UAV: SNR, packet delivery ratio (PDR), long-run reliability, distance, speed and handover state.

Choice of the packet-error curve. The base paper [Shirvanimoghaddam et al., 2022] models packet errors with the normal approximation for finite-length codes: a k-bit packet sent in n channel uses at SNR γ is lost with probability

$$p = Q\left(\frac{n\log_2(1+\gamma) - k + \log_2 n}{\sqrt{n\,V(\gamma)}}\right), \qquad V(\gamma) = \big(1 - (1+\gamma)^{-2}\big)\log_2^2 e. \tag{3.5}$$

Our packets carry $L = 1024$ parameters of 32 bits, so $k = 32{,}768$. Choosing the code rate whose capacity threshold is $\gamma_{\mathrm{th}} = 3\,\mathrm{dB}$, $R = \log_2(1 + 10^{0.3}) = 1.58$ bits per channel use ($n \approx 20{,}704$), places the 50% point of (3.5) at 3 dB, the midpoint of our logistic curve. For a single codeword of this length, however, (3.5) is a waterfall: the packet error rate falls from 90% to 10% within 0.11 dB. The SNR replayed from the trace is one sample per round, while the real SNR varies during a round. If that variation is modelled as a Gaussian spread in dB and (3.5) is averaged over it, a spread of 1.7 dB reproduces the logistic curve with $a = 1$ to within 1.0 percentage point of packet error rate at every SNR (Fig. 3.2). The logistic curve is therefore the base paper's model with unresolved

within-round SNR variation averaged in. The spread is a fitted modelling assumption, not a measured value, and because the 5 and 15 dB anchors are also assumed, the results are best read against the delivery ratios they produce (Section 4.5). Phase II will use (3.5) directly, with the SNR variation taken from the trace.

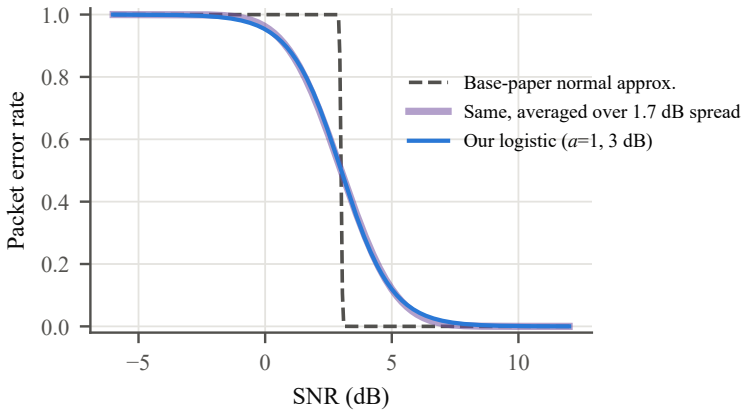


Figure 3.2 Packet error rate against SNR: the base paper's normal approximation for a 32,768-bit packet at rate 1.58 bits per channel use, the same curve averaged over a 1.7 dB Gaussian SNR spread, and the logistic curve used in this work.

3.3 Threat Model

A fixed set of $f = 3$ of the $K = 10$ UAVs is Byzantine. They follow the protocol (they train, transmit on time and use the same channel) but replace their update with one of the following (`fl/attacks.py`, `fl/client.py`):

- **Label flip** (data poisoning): train on labels mapped $c \mapsto 9 - c$.
- **Sign flip**: send $-\lambda \boldsymbol{\Delta}_k$ with $\lambda = 5$.
- **ALIE-style** [Baruch et al., 2019]: send $\boldsymbol{\mu} - z\,\boldsymbol{\sigma} \odot \operatorname{sign}(\boldsymbol{\mu})$ with $z = 1.5$, where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are the mean and standard deviation of the honest updates in the attacker's cell.
- **IPM** [Xie et al., 2019]: send $-\epsilon\boldsymbol{\mu}$ with $\epsilon = 0.5$.

The ALIE and IPM attackers know the honest updates of their cell, which is a strong assumption chosen to stress the defences. The BS's telemetry is assumed authentic; attacks that forge it are left for Phase II.

3.4 Aggregation Rules

All rules receive exactly the same inputs: the received updates, their masks, sample counts and telemetry (`fl/aggregators.py`).

3.4.1 Baselines

FedAvg [McMahan et al., 2017] averages the received updates weighted by n_k, with lost coordinates counted as zeros. **CosAvg**, used only in the preliminary study, is a CosAvg-style baseline inspired by the earlier UAV-FL work of Liao et al. [2023]: it weights each update by $\max(0, \cos(\boldsymbol{\Delta}_k, \bar{\boldsymbol{\Delta}}))\, n_k$, where $\bar{\boldsymbol{\Delta}}$ is the FedAvg mean. It is not an exact reproduction of that paper, whose Algorithm 2 sums pairwise cosine similarities and aggregates the top-ranked updates.

Coordinate-wise median [Yin et al., 2018] and **FLTrust** [Cao et al., 2021] were defined for complete update vectors; we adapt both to the packet masks. The median takes, for each parameter, the median of the delivered values only (the lower middle value when their number is even). FLTrust computes a root update $\mathbf{r}$ on $\mathcal{D}_0$ with as many local steps as the median UAV in the cell, weights each UAV by $\max(0, \cos)$ to $\mathbf{r}$ on its delivered coordinates, rescales the delivered part to the norm of $\mathbf{r}$ on the same coordinates, and averages each coordinate over the UAVs that delivered it. The root-set design is FLTrust's; the mask handling is our adaptation.

Base-paper erasure rules. The base paper [Shirvanimoghaddam et al., 2022] models each upload as erased or received and compares two server rules: average only the fresh updates received in the round (its Eq. (4)), or reuse the last local parameters received from a client whose fresh update is missing (its Eq. (5)). We implement both at packet granularity, which lets them use every delivered packet: *fresh-only* averages each coordinate over the UAVs that delivered it, weighted by n_k; *reuse-old* fills each lost coordinate, and every coordinate of a fully erased upload, with the last parameter value received from that UAV, re-expressed against the current edge model. Neither rule defends against attackers, as the base paper assumes none.

3.4.2 Context-Calibrated Robust Aggregation (CCRA)

CCRA is the aggregator designed in this project. Each BS applies the following steps every round.

1. **Reference update.** Compute $\mathbf{r}$ on $\mathcal{D}_0$, as in FLTrust. Our diagnostics showed that, under non-IID data, the mean or median of the UAV updates is not a reliable reference: attackers and honest UAVs with skewed data had similar cosine similarity to it.

2. **Norm clipping.** Let ρ_k be the RMS of the delivered entries of $\tilde{\mathbf{\Delta}}_k$ and scale the update by $s_k = \min(1,\ \mathrm{med}_j \rho_j/\rho_k)$, with a small constant added to ρ_k so that a zero update is handled. This limits the effect of large-norm updates.

3. **Masked similarity.** $c_k = \cos\big(\mathcal{S}(s_k\tilde{\mathbf{\Delta}}_k; \mathbf{m}_k),\ \mathcal{S}(\mathbf{r}; \mathbf{m}_k)\big)$, where $\mathcal{S}(\cdot; \mathbf{m})$ keeps only the coordinates with $\mathbf{m} = 1$, and $c_k = 0$ for an empty mask or a zero vector. This removes the artifact of lost packets counted as zeros; it does not guarantee that the missing coordinates never bias the comparison.

4. **Behaviour score.** $b_k = \mathrm{clip}\big(c_k/\mathrm{med}_{j:c_j>0} c_j,\ 0,\ 1\big)$. Updates pointing away from the reference get $b_k = 0$, and all scores are 0 if no c_j is positive.

5. **Reputation.** With memory $\beta \in [0, 1)$, initial trust 1 and evidence $e_k \in [0, 1]$,

$$T_k \leftarrow T_k + \eta_k(b_k - T_k), \qquad \eta_k = \begin{cases} (1-\beta)\, e_k, & b_k > T_k, \\ 1-\beta, & b_k \le T_k. \end{cases} \tag{3.6}$$

   Penalties apply at full rate, so an attacker cannot hide by dropping its own packets; only rewards are slowed on poor links.

6. **Gating and averaging.** UAVs with $T_k < \tau$ are excluded. The others get weight $\omega_k = n_k T_k$, discounted for staleness, and each coordinate is averaged only over the UAVs that delivered it (a coordinate with no accepted weight gets a zero update):

$$\mathrm{Agg}_p = \frac{\sum_k \omega_k\, m_{k,p}\, s_k \tilde{\Delta}_{k,p}}{\sum_k \omega_k\, m_{k,p}}. \tag{3.7}$$

The evidence term multiplies five link factors:

$$e_k = \mathrm{PDR}_k \cdot R_k \cdot h_k \cdot \frac{1}{1 + d_k/A} \cdot \frac{1}{1 + |v_k|}, \tag{3.8}$$

namely the packet delivery ratio, long-run reliability, a handover factor ($h_k = 0.5$ during a handover, 1 otherwise), a distance factor with d_k the UAV–BS distance and $A = 800\,\mathrm{m}$ the configured side length of the area, and a factor in the dimensionless normalized speed $|v_k|$.

3.4.3 Ablation Variants

To find out which part of CCRA matters, three variants are compared.

- **CCRA (full)**: all steps, with evidence e_k from telemetry.
- **Mask-only**: $e_k = 1$, but the masks are used in steps 2–6.
- **No-context**: $e_k = 1$ and $\mathbf{m}_k = \mathbf{1}$, so lost coordinates enter the computation as zeros, as a channel-unaware server would see them.

Server-side deadline handling and staleness treatment are shared by all variants; only the mechanisms listed above are ablated, so no-context is not a channel-free system. The code also has an option to correct c_k for the noise expected at the reported SINR. It is inactive in the controlled study, which adds no residual noise to delivered parameters, so the full-versus-mask-only contrast isolates e_k.

3.4.4 Analysis of the Reputation Update

The asymmetric update (3.6) has a consequence that was not intended. Suppose a UAV's evidence is constant, $e_k = e \in (0, 1]$, its scores $b \in [0, 1]$ are i.i.d. and independent of its current reputation, with $0 < \mathbb{E}[b] < 1$, and $0 \le \beta < 1$. Conditional on the current level T, the expected change in reputation is

$$\mathbb{E}[\Delta T \mid T] = (1 - \beta)\, g_e(T), \qquad g_e(T) = e\,\mathbb{E}\big[(b - T)^+\big] - \mathbb{E}\big[(T - b)^+\big], \tag{3.9}$$

so the drift vanishes where

$$e\,\mathbb{E}\big[(b - T^\star)^+\big] = \mathbb{E}\big[(T^\star - b)^+\big]. \tag{3.10}$$

This balance is an expectile condition [Newey and Powell, 1987]: dividing by $1 + e$ gives $q\,\mathbb{E}[(b - T^\star)^+] = (1 - q)\,\mathbb{E}[(T^\star - b)^+]$ with $q = e/(1 + e)$, the first-order condition that defines the q-expectile of b. Our contribution is to apply this relationship to the reputation rule.

Proposition 3.1. Under the assumptions above, g_e has a unique zero $T^\star(e) \in (0, 1)$, the q-expectile of b with $q = e/(1 + e)$. It equals $\mathbb{E}[b]$ when $e = 1$, is nondecreasing in e, and tends to $\operatorname{ess\,inf} b$ as e decreases to zero.

Proof. g_e is continuous with $g_e(0) = e\,\mathbb{E}[b] > 0$ and $g_e(1) = -\mathbb{E}[1 - b] < 0$, so a zero exists in $(0, 1)$. For $T_2 > T_1$, the change $g_e(T_2) - g_e(T_1)$ lies between $-(T_2 - T_1)$ and $-e(T_2 - T_1)$, so g_e is strictly decreasing and the zero is unique. For $e = 1$, $g_1(T) = \mathbb{E}[b] - T$. Increasing e weakly increases $g_e(T)$ for every T, so it cannot lower the zero. For $T < \operatorname{ess\,inf} b$, the downward term vanishes and $g_e(T) > 0$, so the zero is at least $\operatorname{ess\,inf} b$. For any larger T, the downward term $\mathbb{E}[(T - b)^+]$ is strictly positive while the upward term $e\,\mathbb{E}[(b - T)^+]$ vanishes as $e \to 0$, so the limiting zero cannot exceed $\operatorname{ess\,inf} b$. ■

$T^\star$ is the equilibrium of the conditional drift, not a statement that the reputation converges almost surely: with a constant step ($\beta = 0.5$) and random scores it keeps fluctuating, and in general $\mathbb{E}[g_e(T)] \neq g_e(\mathbb{E}[T])$. In the simulator the evidence also changes over time, and scores depend on training, on the other UAVs and on the median normalization, so the constant-evidence, i.i.d. assumptions are an explanatory approximation. Under that approximation, weakening the evidence lowers the equilibrium from the mean score toward the lower end of the score distribution. A finite weak evidence does not put it at the lowest score; that is only the limit as $e \to 0$. An honest UAV with skewed data has noisy scores, so on a weak link this suggests a mechanism by which its trust recovers more slowly and falls below the gate τ more often. An attacker's scores are already near zero, so slowing its rewards adds little. Section 4.5.3 checks whether the measured gate decisions are consistent with this mechanism.

Chapter 4

Experimental Study

This chapter describes how the framework was implemented and verified, the protocol of each experiment, and the results, in the order the work was done: the FedAvg baseline, the preliminary single-seed study, and the controlled 720-run study, followed by a separate 240-run base-paper comparison. Result tables and figures are generated from committed run-level data by the scripts described in Appendix B.

4.1 Implementation and Verification

The framework is written in Python with PyTorch. Runs are configured by a single dataclass, and every run stores its full configuration, per-round history and final model. Automated tests cover data partitioning, the FedAvg update, attack construction, every robust aggregator, the base paper's two erasure rules, the hierarchical server, handover logic, trace loading and the analysis code. A code review of the hierarchical extension found three problems, which were fixed before the controlled study was run: the reputation rule shielded jammed attackers, the no-context ablation still read packet delivery ratios, and FLTrust's root update took too few training steps.

4.2 Experimental Protocol

4.2.1 Preliminary Study

The first comparison used the synthetic jammed channel with one BS, a 30% label-flip attack, 30 rounds and a single seed. It compared FedAvg, the CosAvg-style baseline and an early version of CCRA under four scenarios: no attack or attack, each with an ideal or a jammed lossy channel. Its purpose was to check that the pipeline worked and to estimate effect sizes.

4.2.2 Controlled Ablation Study

The main study used a full factorial design: six aggregators (FedAvg, median, FLTrust and the three CCRA variants), four attack settings (clean, sign flip, ALIE, IPM), three channel conditions (ideal; trace at 15 dB; trace at 5 dB) and 10 seeds, for 720 runs of 100 rounds each with $K = 10$, $B = 3$, $E = 5$ and $\alpha = 0.5$. Within a seed, the data split, initial model, attacker identities and trace positions are identical across methods, so every comparison is paired. There is no jammer and no additive noise on delivered parameters. The training set is the 49,500 client images plus the disjoint 500-image root set. The runs were executed on an NVIDIA A100 GPU.

The analysis plan was fixed before any run:

- The *primary* quantity is the telemetry effect under stress, $[(\text{CCRA}-\text{mask-only})_{5\,\text{dB}} - (\text{CCRA} - \text{mask-only})_{15\,\text{dB}}]$, averaged over the three attacks within each seed.
- An effect counts as practically meaningful only if its 95% interval lies entirely beyond ± 1 percentage point.
- Intervals are percentile bootstraps over seeds with 10,000 resamples; rounds and attack cells are not treated as independent samples.
- A run that diverges numerically is scored at chance accuracy (10%) instead of being dropped, so unstable methods cannot look better by losing their worst runs.
- All other contrasts are exploratory and are not adjusted for multiple comparisons.

4.2.3 Comparison with the Base Paper

The base paper's two erasure rules (Section 3.4.1), fresh-only and reuse-old, are run as a separate comparison on the same seeds, attacker identities, channel conditions and attack settings as the controlled study: $2 \times 4 \times 3 \times 10 = 240$ runs. They are not added to the prespecified matrix, and no parameter is tuned on its seeds. All 240 runs are now complete and validated. Both rules use the same 49,500 client images and reserve the same disjoint 500 root images, although neither uses a trusted reference update.

The follow-up was executed through SwarmCompute on Windows NVIDIA workers. Final attempts used an RTX 2060 for 238 runs and a GTX 1650 for two; 57 final attempts resumed from centrally validated checkpoints. Configurations, source and dataset fingerprints, terminal records and all 120 within-study scientific pairs were checked. Runtime profiles and artifact checksums are retained with the compact results. Hardware and software versions differ from the A100 ablation, so cross-study rankings are descriptive. The paired fresh-only versus reuse-old intervals resample the ten seed pairs within this follow-up; they do not make the two studies a single prespecified experiment.

4.3 FedAvg Baseline Results

The first objective was a working FL pipeline without any defence. Ten UAVs trained TacticalEdgeCNN for 50 rounds with two local epochs, SGD (learning rate 0.01, momentum 0.9, batch size 64), on one BS with an ideal channel. Table 4.1 and Fig. 4.1 show the result.

Table 4.1 FedAvg baseline accuracy after 50 rounds

Data partition	Final test accuracy (%)	Best test accuracy (%)
IID (5,000 balanced images per UAV)	62.62	62.65
Non-IID, Dirichlet $\alpha = 0.5$	60.23	60.58

Non-IID data cost about 2.4 percentage points, and neither curve had fully levelled off by round 50. These accuracies are not direct targets for the later experiments, which use 30 or 100 rounds, one local epoch and, in the controlled study, a hierarchical system; robust methods are instead compared with matched clean controls within each experiment.

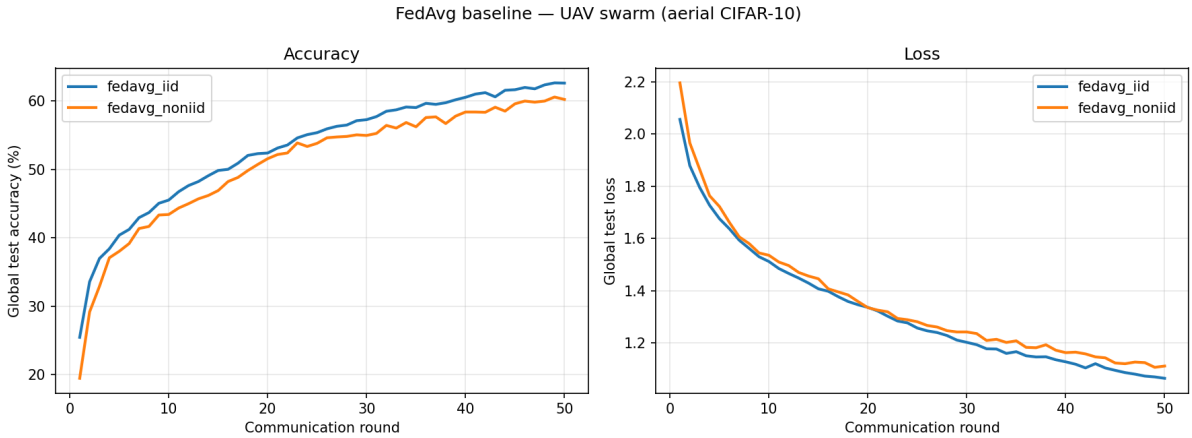


Figure 4.1 Test accuracy and loss of the FedAvg baseline over 50 rounds.

4.4 Preliminary Study Results

The preliminary study compared FedAvg, the CosAvg-style baseline (inspired by Liao et al. [2023], not an exact reproduction) and an early version of CCRA with one BS, 30 rounds, one local epoch and a single seed. Three of the ten UAVs flipped their labels, and in the jammed scenarios a ground jammer targeted three honest UAVs. Table 4.2 lists the final accuracy (mean of the last five rounds), and Figs. 4.2 and 4.3 show the accuracy and the weight each method gave to the attackers.

Table 4.2 Preliminary study: final accuracy (%), single seed, 30 rounds. CosAvg is our CosAvg-style baseline, not the published method.

Scenario	FedAvg	CosAvg	CCRA (early)
No attack, ideal channel	46.5	**47.1**	44.5
No attack, jammed lossy channel	44.6	**45.6**	43.1
30% label flip, ideal channel	37.6	36.6	**43.3**
30% label flip, jammed lossy channel	36.8	39.7	**43.0**

The early CCRA kept about 43% accuracy in every attacked scenario and gave the attackers 1.9% of the aggregation weight on the ideal channel, against 19.1% for CosAvg and 26.0% for FedAvg. It cost about 2.5 points when there was no attack. These results were encouraging, but they could not support a claim:

- they came from one seed, so the size of the random variation was unknown;
- the channel-aware part of CCRA added only 0.8 points over its no-context ablation

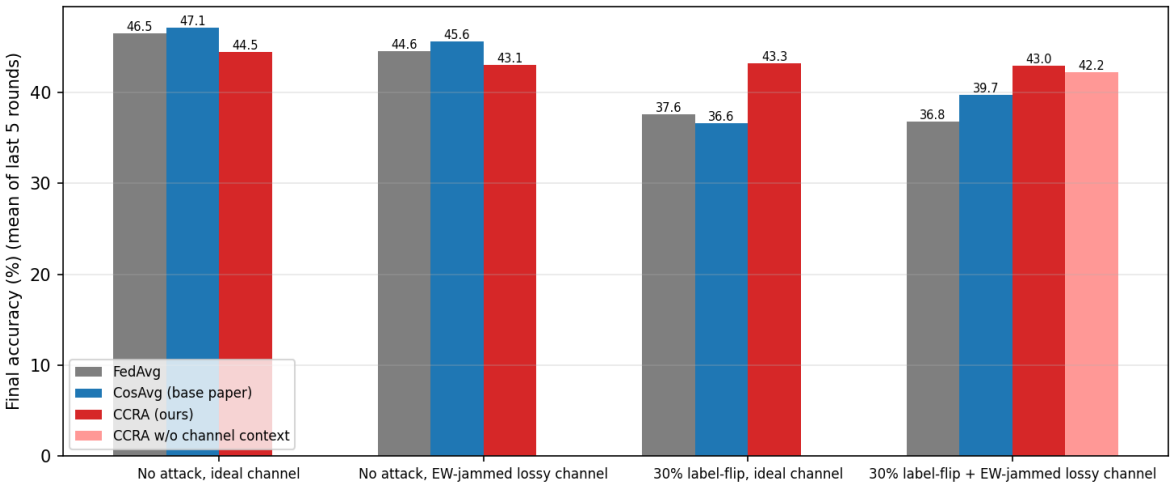


Figure 4.2 Preliminary study: final accuracy in the four scenarios.

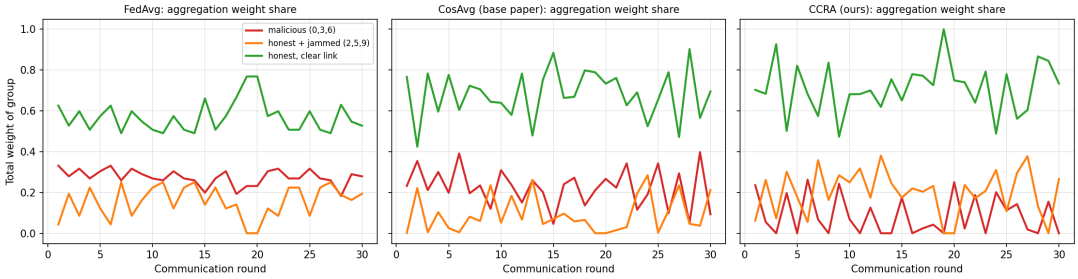


Figure 4.3 Preliminary study: aggregation weight given to attackers and to jammed honest UAVs under label flipping on the jammed channel.

(42.2 to 43.0%), and with one seed the uncertainty in that difference could not be estimated;

- CCRA used a trusted root set while CosAvg did not, so the comparison mixed two differences;
- the reputation rule was revised when the system was extended to hierarchical FL, so the early CCRA is not the exact version evaluated in the controlled study.

These limitations defined the controlled study.

4.5 Controlled Ablation Study Results

All 720 runs finished and were validated. The trace delivered on average 99.8% of packets at 15 dB and 65.0% at 5 dB.

4.5.1 Stability and Overall Accuracy

Table 4.3 summarizes every run, and Fig. 4.4 shows accuracy under attack across the three channel conditions.

Table 4.3 Outcomes over all 720 runs (120 per method). Diverged runs count as 10%.

Method	Diverged	All runs (%)	Attacked, 5 dB (%)
FedAvg	41/120	37.3	29.4
Median	26/120	29.3	21.3
FLTrust	0/120	50.6	**50.1**
No-context	0/120	51.4	45.3
Mask-only	0/120	**52.8**	49.5
CCRA (full)	0/120	51.1	46.0

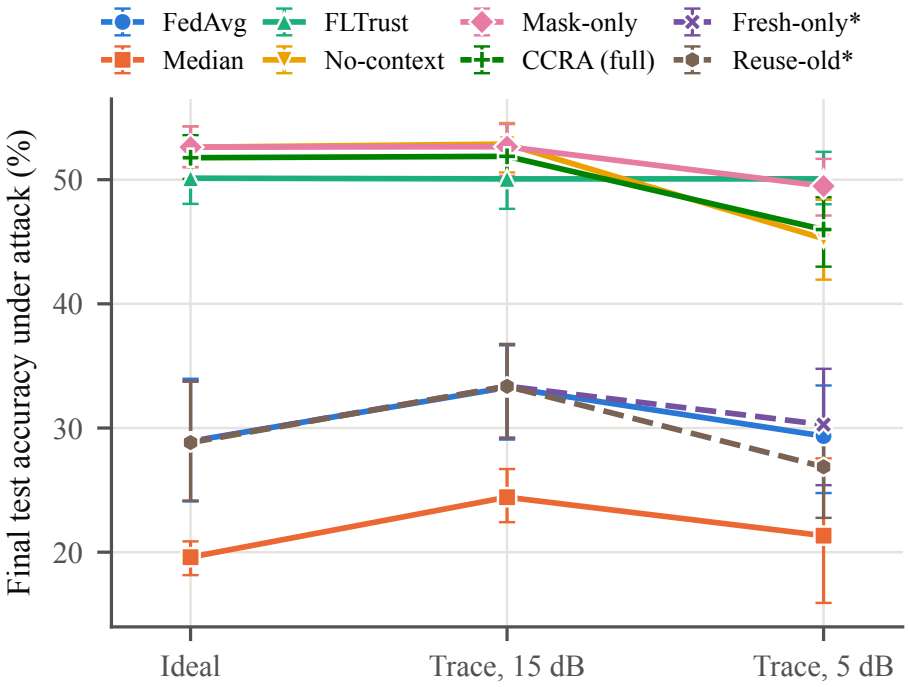


Figure 4.4 Final accuracy under attack (mean over sign flip, ALIE and IPM and 10 seeds). Error bars: 95% seed-bootstrap intervals. Dashed, starred rules are from the separate 240-run follow-up; cross-study rankings are descriptive.

FedAvg diverged in 41 and the coordinate-wise median in 26 of their 120 runs; FedAvg diverged in all 30 sign-flip runs. None of the four methods that use a trusted reference update (FLTrust and the three CCRA variants) diverged in its 120 runs. A plausible explanation for the median's failures, not tested by a separate diagnostic, is that each cell holds only a few UAVs, so one attacker in a cell is already a large fraction of

the values being compared. Over all runs the mask-only variant has the highest accuracy, 52.8%, against 37.3% for FedAvg. Table 4.4 breaks the result down by attack.

Table 4.4 Mean final accuracy (%) by attack, method and channel. Best value in each block and column in bold.

Attack	Method	Ideal	Trace, 15 dB	Trace, 5 dB
No attack	FedAvg	**59.2**	**59.3**	54.1
	Median	52.5	52.7	49.7
	FLTrust	52.5	51.5	52.9
	No-context	57.1	56.7	51.1
	Mask-only	57.1	57.5	**54.7**
	CCRA (full)	55.4	56.6	52.6
Sign flip	FedAvg	10.0	10.0	10.0
	Median	10.0	19.6	12.4
	FLTrust	50.0	49.9	**50.6**
	No-context	**54.2**	**53.8**	46.1
	Mask-only	54.2	53.6	49.9
	CCRA (full)	52.5	52.8	46.5
ALIE	FedAvg	49.8	46.9	42.0
	Median	35.4	28.4	28.1
	FLTrust	49.9	50.6	49.6
	No-context	**52.9**	52.6	46.0
	Mask-only	52.9	**52.7**	**50.8**
	CCRA (full)	52.8	51.9	46.9
IPM	FedAvg	27.0	42.9	36.0
	Median	13.4	25.3	23.5
	FLTrust	50.5	49.7	**50.0**
	No-context	**50.7**	**52.3**	43.6
	Mask-only	50.7	51.7	47.7
	CCRA (full)	50.0	51.0	44.5

Two points in Table 4.4 deserve attention. First, with no attack, plain FedAvg is the most accurate method on the ideal and 15 dB channels; every robust method pays a cost of roughly two to eight points for discarding or down-weighting honest updates. Second, on the 5 dB channel the masked variant is slightly ahead of FedAvg even without an attack, consistent with FedAvg's zero-filled lost coordinates now hurting it.

4.5.2 Which Kind of Channel Awareness Helps

Figure 4.5 shows the paired contrasts that separate the two channel mechanisms.

Erasure masking. On the ideal channel nothing is lost, so mask-only and no-

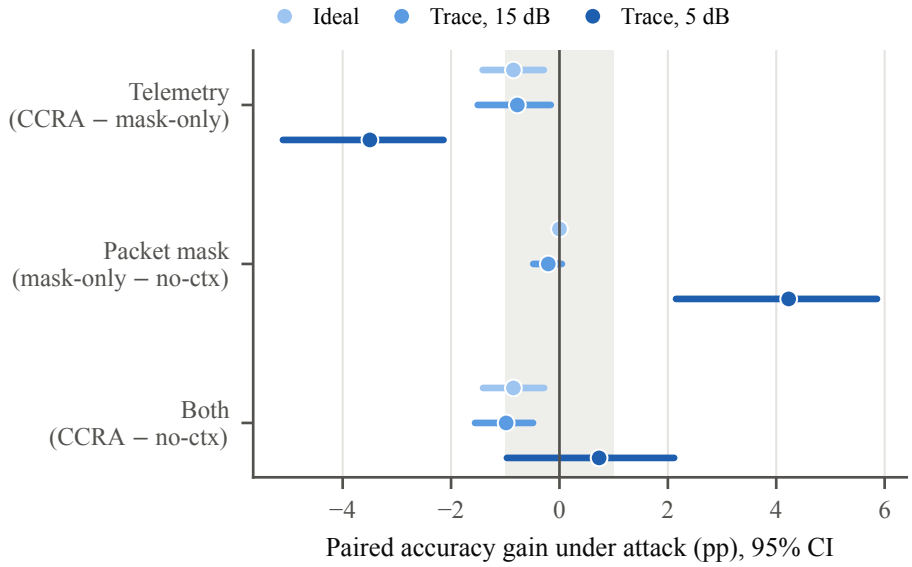


Figure 4.5 Paired accuracy differences under attack with 95% bootstrap intervals. The grey band is the prespecified ±1 point region of practical equivalence.

context are identical. As the link weakens the masks matter more, and at 5 dB masking adds 4.2 points (95% CI [2.2, 5.9]). Treating lost coordinates as zeros shrinks honest updates, distorts their similarity scores and pulls the average towards zero; ignoring them avoids all three effects.

Telemetry in the reputation. The prespecified primary estimate, the telemetry effect at 5 dB minus that at 15 dB, is −2.7 points ([−3.9, −1.5]). The whole interval lies below the −1 point threshold, so by the rule fixed in advance the telemetry mechanism has a meaningful *negative* effect that grows as the link weakens. In absolute terms, at 5 dB CCRA minus mask-only is −3.5 points ([−5.1, −2.1]): 46.0% against 49.5%.

Both together. Full CCRA against no-context combines a gain and a loss; at 5 dB the net effect is small and its interval includes the equivalence region. The earlier impression that "channel context helps" came from this combination, which hid two opposite effects.

4.5.3 Mechanism: Who the Gate Rejects

Section 3.4.4 predicted that slowing rewards on weak links would push honest UAVs' reputations down without doing much to attackers. Figure 4.6 is consistent with that pattern.

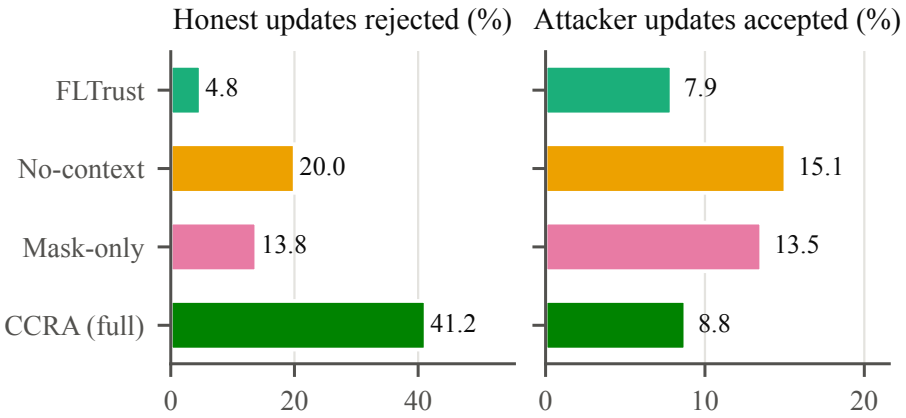


Figure 4.6 Gate decisions under attack at 5 dB, as a percentage of received updates, averaged over attacks and seeds.

With telemetry, the gate rejected 41.2% of honest updates against 13.8% without it, while the share of attacker updates accepted fell only from 13.5% to 8.8%. The accuracy loss from telemetry was accompanied by this rise in honest-update rejection. The rule gave up a large amount of honest data to filter slightly more attacker data, and it did so precisely when honest data were already scarce because of packet loss. These rates are descriptive. They agree with the analysis in Section 3.4.4, but they do not by themselves prove that it is the cause.

4.5.4 Comparison with FLTrust

FLTrust is the closest established method, since it uses the same root set. Its gate rejected only 4.8% of honest updates at 5 dB. It has no reputation memory, since each round is judged on its own, but its norm matching, score normalization, weights and gating also differ from CCRA's, so the lower rejection cannot be attributed to memory alone. In an exploratory paired comparison, mask-only was 2.6 points better than FLTrust under attack at 15 dB ([0.9, 4.5]), while the 5 dB comparison was inconclusive (−0.6 points, [−2.0, 1.0]): the interval supports neither a difference nor practical equivalence, since it is not entirely inside ±1 point. One reading consistent with the mechanism above is that reputation memory helps on good links and stops helping on weak ones.

4.6 Completed Base-Paper Comparison

The 240 terminal outcomes comprise 157 successful runs and 83 numerical divergences. Fresh-only diverged in 41 of 120 runs, and reuse-old in 42 of 120. All 60 sign-flip runs diverged; the remaining 23 divergences occurred under IPM. No clean or ALIE run diverged. These outcomes are retained at 10% final accuracy, rather than excluded or retried into successful runs. All-run accuracy is 37.74% for fresh-only and 36.42% for reuse-old.

Table 4.5 Separate base-paper follow-up: ten paired seeds per cell. Final accuracy includes numerical divergences at chance (10%). Intervals are exploratory paired seed-bootstrap intervals, with no multiplicity adjustment.

Attack	Channel	Fresh (%)	Reuse (%)	Fresh − reuse: pp [95% CI]
No attack	Ideal	59.01	59.01	+0.00 [+0.00, +0.00]
	15 dB	59.44	59.40	+0.04 [−0.17, +0.25]
	5 dB	56.86	51.47	+5.39 [+4.11, +6.74]
Sign flip	Ideal	10.00	10.00	+0.00 [+0.00, +0.00]
	15 dB	10.00	10.00	+0.00 [+0.00, +0.00]
	5 dB	10.00	10.00	+0.00 [+0.00, +0.00]
ALIE	Ideal	49.71	49.54	+0.17 [−0.03, +0.35]
	15 dB	47.21	47.18	+0.03 [−0.34, +0.33]
	5 dB	43.34	39.84	+3.50 [+0.53, +6.53]
IPM	Ideal	26.91	26.95	−0.03 [−0.10, +0.04]
	15 dB	42.90	42.91	−0.00 [−0.12, +0.15]
	5 dB	37.49	30.78	+6.71 [+1.38, +14.97]

Table 4.5 shows identical mean accuracy on the clean ideal channel, where there are no missing packets to reuse. On clean stressed links, fresh-only attains 56.86% versus 51.47% for reuse-old: a gain of 5.39 percentage points (95% CI [4.11, 6.74]). Under attack at 5 dB, the chance-scored means are 30.28% and 26.87%, respectively, a gain of 3.40 points ([0.94, 6.60]). The interval describes a mean over the three attacks within each seed, not 30 independent replicates.

Retaining old parameters does not improve this finite-horizon stressed-channel experiment, and neither rule provides Byzantine robustness. This is compatible with the scope of Shirvanimoghaddam et al. [2022]: its analysis assumes honest clients and whole-upload erasures, whereas the present implementation uses packet erasures in a hierarchical

non-IID CNN task with adversarial updates. These results are an adaptation study, not an exact reproduction or a refutation of the original convergence theorem.

Figure 4.4 now includes both rules alongside the original six methods. The original ablation's findings about masking and telemetry are unchanged.

4.6.1 Our Method against the Base Paper

Both studies use the same seeds, data partitions and attacker identities, so mask-only CCRA can be compared with the base paper's better rule, fresh-only, seed by seed. Under attack at 5 dB, mask-only attains 49.5% against 30.28% for fresh-only, a paired gain of 19.2 percentage points (95% CI [15.9, 23.0]). Without attack at 5 dB, fresh-only is ahead, 56.86% against 54.7%, a difference of -2.1 points ([−2.9, −1.4]). This is the cost of robustness: mask-only judges every update against the root reference and down-weights some honest ones, which helps only when attackers are present. The base paper's rules were designed for honest clients and pay nothing for that protection, but they offer none.

The two studies ran on different hardware (an A100 against RTX 2060 and GTX 1650 workers) and software versions, so these contrasts are cross-study and exploratory. A built-in check bounds the hardware effect. On the ideal channel nothing is erased, so fresh-only reduces exactly to FedAvg, which the controlled study also ran on the same seeds. Across the 24 ideal-channel runs that did not diverge, the two studies agree within 1.1 points per run (mean 0.29 points), and both diverge in exactly the same runs. That is far smaller than the 19.2-point gap under attack.

Operational interruptions were kept separate from numerical divergence. Clean checkpointed stops that had been mislabeled as application failures were requeued after checkpoint verification; Windows progress-file permission failures received bounded retries with unchanged scientific configurations. Failed attempts remain in the coordinator audit history. The final validation contains one accepted terminal result per planned job, with no missing jobs. Recovery preserves saved state and random generators but does not establish bitwise reproducibility across GPUs.

4.7 Discussion

4.7.1 What the Results Show

1. All four root-referenced methods avoided divergence in their 120 runs each, while FedAvg and the median diverged in some runs. This association does not isolate the contribution of the root set.

2. Among channel-aware mechanisms, erasure masking is simple; under attack its 15 dB accuracy contrast lies within the prespecified equivalence region, and it helps substantially on a weak link. No computational or communication overhead was measured.

3. The intuitive idea of letting link quality soften judgement backfired. The analysis of Section 3.4.4 suggests a structural reason: an asymmetric reputation update can bias trust downwards for the UAVs it was meant to protect.

4. The separate base-paper comparison shows that fresh-only averaging can exceed reuse-old under stressed packet loss. Both erasure rules diverged under sign flip, illustrating why erasure handling alone is insufficient for this threat model.

4.7.2 Why a Negative Result Is Useful

The preliminary study suggested that CCRA's channel context was its key ingredient. Had that claim been carried forward, Phase II would have been built on a component that harms accuracy. The prespecified design, paired seeds and chance scoring of diverged runs made it possible to reject the claim with confidence, and the analysis in Section 3.4.4 points to the specific rule that must change.

4.7.3 Threats to Validity

- **Channel realism.** The trace comes from one measured BS; the other BSs are virtual, and the 5 and 15 dB anchors are assumptions, not calibrated values. The runs contain few handovers, so no claim is made about high mobility.

- **Scope of learning task.** One dataset (aerial-transformed CIFAR-10, not genuine UAV imagery), one model, ten UAVs and fixed attack strengths were used.
- **Trust assumptions.** The root set is assumed clean; no trusted execution environment or attestation was implemented or evaluated.
- **Attacker model.** Telemetry is assumed authentic, and adaptive attacks on the reputation itself [Liu et al., 2026] were not tested.
- **Statistics.** Ten seeds give approximate intervals; secondary contrasts are not adjusted for multiple comparisons. Any revised rule must be tested on new seeds, not re-tuned on these ten.
- **Execution environment.** The A100 ablation and Windows-worker follow-up used different GPU/software profiles and source fingerprints. Checkpoint recovery was verified, but hardware-controlled cross-study effects and runtime speedups were not measured.

Chapter 5

Conclusions and Future Scope of Study

5.1 Summary of the Work

This project built a simulation framework for federated learning over a UAV swarm whose updates travel over lossy air-to-ground links and in which some UAVs are Byzantine. The framework supports hierarchical aggregation at three base stations, handover, packet erasures from a synthetic jammed channel or a measured UAV trace, five attacks and twelve selectable aggregation rules (FedAvg, the CosAvg-style baseline, median, trimmed mean, Krum, Multi-Krum, FLTrust, three CCRA variants and the base paper's two erasure rules), with automated tests. The controlled study evaluated six methods; a separate 240-run follow-up evaluated the two base-paper rules. A FedAvg baseline was established first. A reputation-based aggregator, CCRA, was then designed with two forms of channel awareness, erasure masking and telemetry-weighted reputation. After a promising but inconclusive single-seed study, a prespecified, paired study of 720 runs separated the two forms and measured each one's effect. The results have been written up as a manuscript in IEEE letter format.

5.2 Conclusions

1. Non-IID data reduced FedAvg accuracy from 62.6% to 60.2% in the baseline, and a 30% label-flip attack reduced it by about ten points in the preliminary study.

2. None of the four aggregators that judge updates against a trusted reference update

diverged in its 120 runs (480 runs combined), whereas FedAvg and the coordinate-wise median diverged in 41 and 26 of 120 runs each.

3. Ignoring lost coordinates instead of treating them as zeros (erasure masking) improved accuracy under attack on a 5 dB link by 4.2 percentage points (95% CI [2.2, 5.9]).

4. Using link telemetry to slow reputation rewards had a meaningful negative effect: the prespecified primary estimate was -2.7 points ($[-3.9, -1.5]$), accompanied by a rise in honest-update rejection from 13.8% to 41.2%.

5. A plausible cause is structural. With slowed rewards but full-rate penalties, the constant-evidence drift analysis (3.10) shows that weaker evidence lowers the equilibrium of an honest UAV's reputation from its mean score toward the lower tail of its scores, and the measured gate decisions are consistent with this mechanism.

6. On weak links, the exploratory comparison of the best variant with FLTrust was inconclusive; a clear advantage over FLTrust appeared only at 15 dB.

7. Against the base paper, mask-only CCRA was 19.2 points more accurate than its better erasure rule under attack at 5 dB ([15.9, 23.0]), and -2.1 points less accurate without attack ($[-2.9, -1.4]$): robustness to attackers costs a little accuracy when there are none.

8. Within the base paper's rules, fresh-only exceeds reuse-old by 5.39 points on clean stressed links ([4.11, 6.74]). The rules diverged in 41/120 and 42/120 runs; neither survived the tested sign-flip attack. This does not contradict the original paper's honest-client convergence analysis.

5.3 Scope for Future Work (Phase II)

1. **Extend the completed base-paper comparison.** Repeat key contrasts on a common hardware/software stack and fresh seeds. Evaluate whether filling lost packets from a trusted UAV's previous parameters can improve on masking without retaining poisoned values.

2. **Redesign the reward rule** so that weak links do not bias reputation downwards, for example by making the update symmetric in the evidence, or by normalizing each UAV's score against its own recent history, and freeze the rule before testing.

3. **Confirm on fresh seeds and conditions.** Evaluate the revised rule on new seeds and channel settings rather than the ten used here, so that tuning cannot inflate the result.

4. **Mobility and handover.** Use traces or mobility models with frequent handovers, and more base stations, to test whether telemetry helps when the dominant impairment is handover rather than fading.

5. **Stronger attackers.** Add adaptive attacks that target the reputation itself and attackers that spoof their link telemetry.

6. **Network-level simulation.** Replace the packet-error model with a network simulator such as NS-3 to capture medium access, retransmissions and latency.

7. **Generalization.** Repeat the key comparisons with another dataset, a larger model and a larger swarm, and submit the manuscript.

References

M. S. H. Abad, E. Ozfatura, D. Gündüz, and Ö. Erçetin. Hierarchical federated learning across heterogeneous cellular networks. In *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, pages 8866–8870, 2020. doi: 10.1109/ICASSP40776.2020.9054634.

A. Alkhateeb, G. Charan, T. Osman, A. Hredzak, J. Morais, U. Demirhan, and N. Srinivas. DeepSense 6G: A large-scale real-world multi-modal sensing and communication dataset. *IEEE Commun. Mag.*, 61(9):122–128, 2023. doi: 10.1109/MCOM.006.2200730.

S. Barkatsa, M. Diamanti, P. Charatsaris, E. E. Tsiropoulou, and S. Papavassiliou. Jamming attack mitigation in wireless federated learning networks using Bayesian games. *IEEE Netw. Lett.*, 6(4):247–251, 2024. doi: 10.1109/LNET.2024.3499360.

M. Baruch, G. Baruch, and Y. Goldberg. A little is enough: Circumventing defenses for distributed learning. In *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.

P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer. Machine learning with adversaries: Byzantine tolerant gradient descent. In *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.

X. Cao, M. Fang, J. Liu, and N. Z. Gong. FLTrust: Byzantine-robust federated learning via trust bootstrapping. In *Proc. Netw. Distrib. Syst. Secur. Symp. (NDSS)*, 2021. doi: 10.14722/ndss.2021.24434.

G. Charan and A. Alkhateeb. Sensing-aided 6G drone communications: Real-world datasets and demonstration. *IEEE Open J. Commun. Soc.*, 6:8745–8774, 2025. doi: 10.1109/OJCOMS.2025.3612320.

X. Fan, G. Li, C. Luo, Y. Hong, Z. Chen, and T. Chen. UAV-assisted heterogeneous

federated learning over the air against Byzantine attacks. *Tsinghua Sci. Technol.*, 31 (2):904–919, 2026. doi: 10.26599/TST.2024.9010167.

T.-M. H. Hsu, H. Qi, and M. Brown. Measuring the effects of non-identical data distribution for federated visual classification. arXiv:1909.06335, 2019.

S. P. Karimireddy, L. He, and M. Jaggi. Learning from history for Byzantine robust optimization. In *Proc. Int. Conf. Mach. Learn. (ICML)*, volume 139 of *PMLR*, pages 5311–5319, 2021.

S. P. Karimireddy, L. He, and M. Jaggi. Byzantine-robust learning on heterogeneous datasets via bucketing. In *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

M. Li, D. Xiao, J. Liang, and H. Huang. Communication-efficient and Byzantine-robust differentially private federated learning. *IEEE Commun. Lett.*, 26(8):1725–1729, 2022. doi: 10.1109/LCOMM.2022.3180113.

J. Liao, B. Jiang, P. Zhao, L. Ning, and L. Chen. Unmanned aerial vehicle-assisted federated learning method based on a trusted execution environment. *Electronics*, 12 (18):3938, 2023. doi: 10.3390/electronics12183938.

X. Liu, E. Liu, W. Ni, H. Zhang, Z. Wei, and E. Hossain. Learning to evade: Adaptive and stealthy attack on reputation-based federated learning. *IEEE Commun. Lett.*, 30: 1836–1840, 2026. doi: 10.1109/LCOMM.2026.3685107.

B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proc. Int. Conf. Artif. Intell. Statist. (AISTATS)*, volume 54 of *PMLR*, pages 1273–1282, 2017.

W. K. Newey and J. L. Powell. Asymmetric least squares estimation and testing. *Econometrica*, 55(4):819–847, 1987. doi: 10.2307/1911031.

K. Pillutla, S. M. Kakade, and Z. Harchaoui. Robust aggregation for federated learning. *IEEE Trans. Signal Process.*, 70:1142–1154, 2022. doi: 10.1109/TSP.2022.3153135.

M. Shirvanimoghaddam, A. Salari, Y. Gao, and A. Guha. Federated learning with erroneous communication links. *IEEE Commun. Lett.*, 26(6):1293–1297, 2022. doi: 10.1109/LCOMM.2022.3167094.

H. Sifaou and G. Y. Li. Robust federated learning via over-the-air computation. In *Proc. IEEE Int. Workshop Mach. Learn. Signal Process. (MLSP)*, pages 1–6, 2022. doi: 10.1109/MLSP55214.2022.9943401.

Y. Xiao, Y. Ye, S. Huang, L. Hao, Z. Ma, M. Xiao, S. Mumtaz, and O. A. Dobre. Fully decentralized federated learning-based on-board mission for UAV swarm system. *IEEE Commun. Lett.*, 25(10):3296–3300, 2021. doi: 10.1109/LCOMM.2021.3095362.

C. Xie, O. Koyejo, and I. Gupta. Fall of empires: Breaking Byzantine-tolerant SGD by inner product manipulation. In *Proc. Conf. Uncertainty Artif. Intell. (UAI)*, volume 115 of *PMLR*, pages 261–270, 2019.

J. Yao, W. Shi, W. Xu, Z. Yang, A. L. Swindlehurst, and D. Niyato. Byzantine-resilient over-the-air federated learning under zero-trust architecture. *IEEE J. Sel. Areas Commun.*, 43(6):1954–1969, 2025. doi: 10.1109/JSAC.2025.3560046.

D. Yin, Y. Chen, K. Ramchandran, and P. Bartlett. Byzantine-robust distributed learning: Towards optimal statistical rates. In *Proc. Int. Conf. Mach. Learn. (ICML)*, volume 80 of *PMLR*, pages 5650–5659, 2018.

Experimental Settings

Table A.1 Settings of the controlled study

Parameter	Value
UAVs K, base stations B	10, 3
Byzantine UAVs f	3, fixed per seed across all methods
Rounds; synchronization period E	100; 5
Local training	1 epoch, SGD, learning rate 0.01, momentum 0.9, batch 64, weight decay 5×10^{-4}
Model	TacticalEdgeCNN, 156,074 trainable parameters, no normalization layers
Data	Aerial-transformed CIFAR-10, Dirichlet $\alpha = 0.5$, 49,500 training images to UAVs
Root set $\mathcal{D}_0$	500 images (50 per class), disjoint from all UAV data
Reputation	$\beta = 0.5$, gate $\tau = 0.3$, initial trust 1
Staleness discount	$\exp(-0.5 \cdot \text{staleness})$; late updates dropped
Packets	$L = 1024$ parameters, 32 bits per parameter
Packet errors	logistic, slope $a = 1$, threshold $\gamma_{\text{th}} = 3\,\text{dB}$
Round deadline; bandwidth	2 s; 20 MHz
Handover	3 dB hysteresis, 2-round time to trigger
Trace	DeepSense 6G Scenario 23, stride 5 samples per round, segments ≥ 200 samples
Reference SNR $\bar{\gamma}$	assumed anchors: 15 dB (good link), 5 dB (weak link)
Jammer; parameter noise	none; none
Attacks	sign flip $\lambda = 5$; ALIE-style $z = 1.5$; IPM $\epsilon = 0.5$
Seeds	41–50

Reproducing the Results

All code, the run-level summary results and this report are in the repository https://github.com/sasank-555/uav-swarm-fl. The raw DeepSense 6G data are access-controlled and must be obtained from the dataset provider; per-run model checkpoints are kept locally and are not distributed. The training-source commit pinned for the controlled study is recorded in `results/CONTEXT_STUDY_FINAL.md`.

Table B.1 Repository layout

Path	Contents
`fl/`	Simulator: data, model, client, channel, topology, attacks, aggregators, server
`tests/`	Automated simulator, coordinator and analysis tests
`run_context_study.py`	Plans and runs the 720-run controlled study
`analyze_context_study.py`	Prespecified analysis with paired bootstrap
`results/context_study/`	Run-level CSV and analysis of the controlled study
`results/base_paper_comparison/`	Compact evidence, paired estimates and validation of the 240-run follow-up
`scripts/make_paper_figures.py`	Builds every figure, table and number in the paper and this report
`paper/`	Manuscript in IEEE letter format
`report/`	This report

The main steps are:

1. Install the dependencies: `pip install -r requirements.txt`.
2. Prepare CIFAR-10: `python prepare_data.py`.
3. Convert the DeepSense Scenario 23 download:
   `python scripts/prepare_deepsense23.py /path/to/Scenario23 -out data/channel_traces/deepsense23.npz`.

4. Run the study (720 GPU runs): `python run_context_study.py -profile final -device cuda -parallel 4`.
5. Analyse it: `python analyze_context_study.py results/context_final`.
6. Rebuild all figures and numbers: `python scripts/make_paper_figures.py`.
7. Build the documents: in `paper/` and in `report/`, run `tectonic -X compile main.tex`, or pdfLaTeX with BibTeX.

Steps 6 and 7 alone regenerate the figures and documents from the committed results. The completed base-paper follow-up uses frozen training revision `ad06055` and the same runner with `-aggs bp_fresh bp_reuse`. Its full checkpoints, coordinator audit and prepared inputs remain local. Compact publication checks source/data fingerprints, terminal histories and all 120 paired client-partition summaries. Recompute the follow-up intervals from the committed summary data without training:

```
python scripts/analyze_base_paper.py -csv
results/base_paper_comparison/base_paper_runs.csv
```

Then repeat steps 6 and 7 above. The result CSV records the final attempt's runtime profile and resumed round; an attempt that resumed may include earlier work from a different device. The source fingerprints of both studies are retained separately.